Weighted Monte Carlo

Marius P. Furter

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1 Background

The following definitions and results are included for reference.

Notation and conventions. We write $f \mathbin{⨾} g$ for composition in *diagrammatic order*: first f, then g. We use $[\![\phi]\!]$ to denote the *Iverson bracket*: $[\![\phi]\!] = 1$ if ϕ is true and $[\![\phi]\!] = 0$ otherwise. We denote by Stoch the category of standard Borel spaces and probability kernels, and by sKern the category of standard Borel spaces and s-finite kernels. A *CD-category* (copy-delete category) is a symmetric monoidal category in which every object X carries a commutative comonoid structure, given by a copy map $\mathsf{cp}_X \colon X \to X \otimes X$ and a delete map $\mathsf{del}_X \colon X \to I$. We write $\mathcal{C}^{\mathsf{del}}$ for the wide subcategory of arrows that commute with del. We write $\|s\|_\infty := \sup_x |s(x)|$ for the supremum norm.

Definition 1.1 (Kernels)**.** Let (X, Σ_X) and (Y, Σ_Y) be measurable spaces. A *kernel* $f \colon X \rightsquigarrow Y$ is a function $f \colon \Sigma_Y \times X \to \overline{\mathbb{R}}_{\geq 0}$ such that

(a) for each $x \in X$, the map $B \mapsto f(B \mid x)$ is a measure,
(b) for each $B \in \Sigma_Y$, the function $x \mapsto f(B \mid x)$ is measurable.

A kernel $f \colon X \rightsquigarrow Y$ is called *finite* if there is a finite uniform bound $r \in \mathbb{R}_{\geq 0}$ with $f(Y \mid x) \leq r$ for all $x \in X$. We call f *s-finite* if it can be written as a countable sum of finite kernels $f_i \colon X \rightsquigarrow Y$:

$$f(B \mid x) = \sum_{i=1}^{\infty} f_i(B \mid x),$$

for all $x \in X$ and $B \in \Sigma_Y$. In this case, we write $f = \sum_i f_i$ and call it a finite decomposition.

It is highly useful to view kernels as operating on measurable functions $t \colon X \to \overline{\mathbb{R}}_{\geq 0}$, which we denote by $\mathfrak{M}^+(X)$.

Definition 1.2 (Kernel operator)**.** A function $\mathbb{E}_f \colon \mathfrak{M}^+(Y) \to \mathfrak{M}^+(X)$ is called a *kernel operator* if it satisfies

(a) (Linearity) For $\alpha, \beta \in \mathbb{R}_{\geq 0}$ and $s, t \in \mathfrak{M}^+(Y)$, $\mathbb{E}_f(\alpha s + \beta t) = \alpha\, \mathbb{E}_f(s) + \beta\, \mathbb{E}_f(t)$.

(b) (Monotonicity) If $s, t \in \mathfrak{M}^+(X)$, and $s \leq t$ everywhere, then $\mathbb{E}_f(s) \leq \mathbb{E}_f(t)$.
(c) (Monotone convergence) If t_n is a non-decreasing sequence in $\mathfrak{M}^+(Y)$, then $\lim_n \mathbb{E}_f(t_n) = \mathbb{E}_f(\lim t_n)$.

Moreover, we call $\mathbb{E}_f$ *finite* if there is $r \in \mathbb{R}_{\geq 0}$ with $\mathbb{E}_f(1) < r$ where 1 denotes the constant function. We call $\mathbb{E}_f$ *s-finite* if there is a sequence of finite kernel operators $\mathbb{E}_{f_i}$ with $\mathbb{E}_f = \sum_{i=1}^{\infty} \mathbb{E}_{f_i}$.

Theorem 1.3. *There is a one-to-one correspondence between kernels $f\colon X \rightsquigarrow Y$ and kernel operators $\mathbb{E}_f\colon \mathfrak{M}^+(Y) \to \mathfrak{M}^+(X)$. A kernel f is sent to the operator*

$$\mathbb{E}_f(t)(x) := \int_Y t(y)\, f(dy \mid x).$$

Conversely, each operator $\mathbb{E}_f$ is sent to the kernel

$$f(B \mid x) := \mathbb{E}_f([\![y \in B]\!])(x).$$

Moreover, this correspondence restricts to s-finite kernels and s-finite kernel operators. We will denote $\mathbb{E}_f[t(y) \mid x] := \mathbb{E}_f(t)(x)$.

As noted by Staton 2017 and Cho and Jacobs 2019, s-finite kernels assemble into a CD-category.

Theorem 1.4. *The category* sKern *of standard Borel spaces and s-finite kernels is a CD-category.*

The following method of constructing CD-categories is described in Fritz 2020.

Definition 1.5 (Symmetric monoidal monad)**.** A monad M is called *symmetric monoidal* if it comes equipped with a natural transformation

$$\nabla_{X,Y}\colon MX \otimes MY \to M(X \otimes Y)$$

that makes M a symmetric monoidal functor with unit $\eta_I\colon I \to MI$, and such that η and μ are monoidal transformations. Moreover, we call M *affine* if $MI \cong I$.

Proposition 1.6. *The Kleisli category Kl_M of a symmetric monoidal monad $M\colon \mathcal{C} \to \mathcal{C}$ is a symmetric monoidal category (SMC). The monoidal product of $f_i\colon X_i \to MY_i$ is defined by*

$$X_1 \otimes X_2 \xrightarrow{f_1 \otimes f_2} MY_1 \otimes MY_2 \xrightarrow{\nabla} M(Y_1 \otimes Y_2).$$

Moreover, the identity-on-objects functor $L\colon \mathcal{C} \to \mathsf{Kl}_M$ that sends $f \mapsto f \mathbin{⨾} \eta$ is symmetric strict monoidal.

Corollary 1.7. *Let M be a symmetric monoidal monad on a CD-category $\mathcal{C}$. Then Kl_M is again a CD-category. Moreover, if $\mathcal{C}$ is a Markov category and $MI \cong I$, then Kl_M is a Markov category as well.*

2 Weighted Monte Carlo

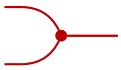
2.1 Weighted kernels

2.1.1 Weights monoid

Definition 2.1 (Weights monoid)**.** Let $\mathcal{C}$ be an SMC. A *monoid object in* $\mathcal{C}$ is an object W, along with

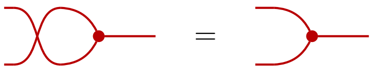
(a) a multiplication arrow $m\colon W \otimes W \to W$,
(b) a unit arrow $e\colon I \to W$,

depicted as:

 $:= m$ $:= e$

satisfying associativity and unitality conditions:

Moreover, W is called *commutative* if



From now on, we will call a commutative monoid object a *weights monoid*.

Proposition 2.2. *Let (W, m, e) be a weights monoid in an SMC $\mathcal{C}$. Then the functor $(-\otimes W)\colon \mathcal{C} \to \mathcal{C}$ is a symmetric monoidal monad (Definition 1.5) with*

$$\mu_X\colon (X\otimes W)\otimes W \xrightarrow{\alpha} X\otimes(W\otimes W) \xrightarrow{X\otimes m} X\otimes W,$$
$$\eta_X\colon X \xrightarrow{\rho^{-1}} X\otimes I \xrightarrow{X\otimes e} X\otimes W,$$
$$\nabla_{X,Y}\colon (X\otimes W)\otimes(Y\otimes W) \xrightarrow{s} (X\otimes Y)\otimes(W\otimes W) \xrightarrow{(X\otimes Y)\otimes m} (X\otimes Y)\otimes W,$$

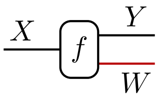
where s is the unique coherence arrow composed of σ and α.

Proof. This is a standard result (see e.g. Brandenburg 2014, Ex. 6.3.12). ∎

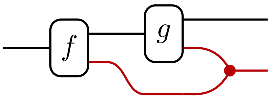
We can now invoke Proposition 1.6 and Corollary 1.7 to construct new CD-categories as the Kleisli category of $(-\otimes W)$.

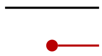
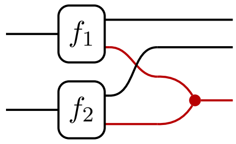
Corollary 2.3. *Let $\mathcal{C}$ be an SMC. Then the Kleisli category $\mathcal{C}_W := \mathsf{Kl}_{(-\otimes W)}$ is an SMC with a symmetric strict monoidal identity-on-objects functor $e\colon \mathcal{C} \to \mathcal{C}_W$ sending $f \mapsto \rho^{-1} ⨾ (f\otimes e)$. Moreover, if $\mathcal{C}$ is a CD-category, then so is $\mathcal{C}_W$.*

A ***weighted arrow*** $f\colon X \to Y$ in $\mathcal{C}_W$ is a $\mathcal{C}$-arrow $f\colon X \to Y\otimes W$. We draw the string representing the weights in red:

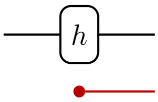


Kleisli composition, identities and monoidal products can then be written as follows:

$f \,⨾\, g \; =$ 

id $=$ 
$f_1 \otimes f_2 \; =$ 

Moreover, any $\mathcal{C}$-arrow $h\colon X \to Y$ can be thought of as a weighted arrow by assigning its output weight e. This is precisely the action of the inclusion functor $e\colon \mathcal{C} \to \mathcal{C}_W$.



We can think of the Kleisli composition as the following sampling procedure:

```
let (y, wf) = f(x) in    # draw (y, w_f) ~ f(- | x)
let (z, wg) = g(y) in    # draw (z, w_g) ~ g(- | y)
(z, wf*wg)               # multiply weights
```

Similarly, the monoidal product corresponds procedurally to

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let (y1, w1) = f1(x1) in    # draw (y_1, w_1) ~ f_1(- | x_1)
let (y2, w2) = g2(x2) in    # draw (y_2, w_2) ~ f_2(- | x_2)
((y1,y2), w1*w2)            # multiply weights
```

Hence, the category $\mathcal{C}_W$ implements the usual way of handling weighted samples.

2.1.2 The CD-category of weighted s-finite kernels

We are interested primarily in sKern_W, whose arrows are ***weighted s-finite kernels***. We will use either ***raw weights*** $W^r := (\overline{\mathbb{R}}_{\geq 0}, \cdot, 1)$ or ***log-weights*** $W^l := (\overline{\mathbb{R}}, +, 0)$. Both options are equivalent under the isomorphism of monoids sending $w \mapsto \log(w)$, with the convention that $\log(0) = -\infty$ and $\log(\infty) = \infty$. If we do not say otherwise, we will work with raw weights W^r. In software implementations, however, log-weights W^l are preferable since they are more numerically stable.

We now explicitly describe the structure sKern_W. Its objects are standard Borel spaces and its arrows are s-finite weighted kernels $f\colon X \rightsquigarrow Y \otimes W$ (Definition 1.1). Hence, for each $x \in X$, a weighted kernel returns a joint measure over outputs Y and weights W. Given $g\colon Y \rightsquigarrow Z \otimes W$, the composition $f \,⨾\, g$ acts on measurable functions $t \in \mathfrak{M}^+(Z \otimes W)$ by

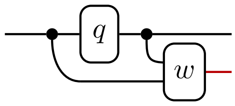
$$\int_Z \int_{W_f} \int_{W_g} t(z, w_f \cdot w_g) \int_Y g(dz, dw_g \mid y)\, f(dy, dw_f \mid x).$$

The Kleisli identity $\mathsf{id}_X^{\mathsf{Kl}} \colon X \rightsquigarrow X \otimes W$ is the probability kernel which acts on $t \in \mathfrak{M}^+(X \otimes W)$ by $t(x, 1)$. The monoidal product of $f_1 \colon X_1 \rightsquigarrow Y_1 \otimes W$ and $f_2 \colon X_2 \rightsquigarrow Y_2 \otimes W$ acts on $t \in \mathfrak{M}^+(Y_1 \otimes Y_2 \otimes W)$ according to

$$\int_{Y_1} \int_{Y_2} \int_{W_1} \int_{W_2} t(y_1, y_2, w_1 \cdot w_2)\, f_1(dy_1, dw_1 \mid x_1)\, f_2(dy_2, dw_2 \mid x_2).$$

Finally, any s-finite kernel $f \colon X \rightsquigarrow Y$ lifts to the weighted kernel $e(f) \colon X \rightsquigarrow Y \otimes W$ which acts on measurable functions $t \in \mathfrak{M}^+(Y \otimes W)$ by $\int_Y t(y, 1)\, f(dy \mid x)$.

A common way to construct weighted kernels $k \colon X \rightsquigarrow Y \otimes W$ is by weighting a kernel $q \colon X \rightsquigarrow Y$ by a deterministic function $w \colon X \times Y \to W$:



The result acts by $t(y, w) \mapsto \int_Y t(y, w(x, y))\, q(dy \mid x)$.

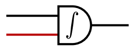
2.1.3 Integrating weights

This section describes a method of constructing functors $\mathcal{C}_W \to \mathcal{C}$ based on intuitions about weighted kernels. Any weighted kernel $f \colon X \rightsquigarrow Y \otimes W$ can be converted to an unweighted kernel $\bar{f} \colon X \rightsquigarrow Y$ by averaging over the weights:

$$\bar{f}(A \mid x) := \int_W w\, f(A, dw \mid x).$$

This operation can be realized by composing f with the following special kernel.

Definition 2.4 (Integrator kernel)**.** For any space Y, the *integrator kernel* $\int_Y \colon Y \otimes W \rightsquigarrow Y$ is defined by $\int_Y(A \mid y, w) := w\, [\![y \in A]\!]$. In string diagrams, we draw integrators as follows:

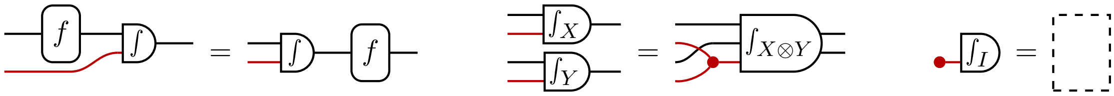


It acts on $t \in \mathfrak{M}^+(Y)$ by $t(y) \mapsto w \cdot t(y)$. Composing $f \colon X \rightsquigarrow Y \otimes W$ with the integrator kernel yields a weighted average:

$$(f \,\fatsemi \textstyle\int_Y)(A \mid x) = \displaystyle\int_Y \int_W w\, [\![y \in A]\!]\, f(dy, dw \mid x) = \int_W w\, f(A, dw \mid x).$$

We now describe the abstract properties of integrator kernels that will allow us to transfer the concept to other categories. First, each integrator forms a ***monad algebra*** over $(- \otimes W)$. In string diagrams, this means the following *unit* and *action* laws hold:

Second, the integrators on different objects assemble into a ***monoidal transformation***. This entails the following compatibility with the SMC structure:

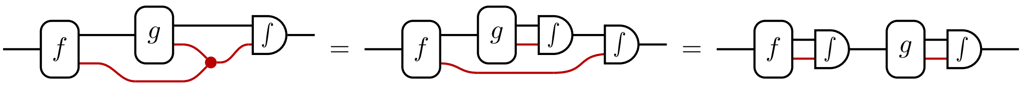


We use these properties to define abstract integrators in any SMC.

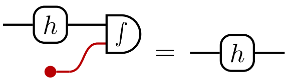
Definition 2.5. Let W be a weights monoid in an SMC $\mathcal{C}$. A *supply of integrators* for W is a monoidal transformation $\int : (- \otimes W) \Rightarrow \mathsf{Id}$ such that every component $\int_X : X \otimes W \to X$ is an $(- \otimes W)$ algebra.

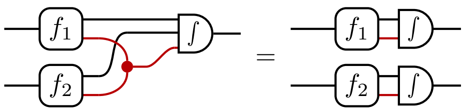
Proposition 2.6. *A supply of integrators induces a symmetric strict monoidal functor $\int_W : \mathcal{C}_W \to \mathcal{C}$ that sends $f \mapsto f \mathbin{⨾} \int$. Moreover, the functor $e \colon \mathcal{C} \to \mathcal{C}_W$ from Corollary 2.3 is a section of $\int_W$.*

Proof. We first check that $\int_W$ is a functor. It preserves identities due to the monad algebra unit law. It preserves composition by applying the action law, followed by naturality with $(g \mathbin{⨾} \int)$:



It is a section of $e \colon \mathcal{C} \to \mathcal{C}_W$ due to the algebra unit law (below left). We now verify that $\int_W$ is symmetric strict monoidal. Since it is identity-on-objects by definition, we only need to check that it strictly preserves the monoidal products and coherence arrows. The former follows from the monoidal transformation law (below right):





The coherence arrows in $\mathcal{C}_W$ are of the form $e(\alpha)$. Since $\int_W$ is a section, $\int_W(e(\alpha)) = \alpha$ shows that they are preserved, along with any other structures (e.g. copy and delete) lifted in this way. ■

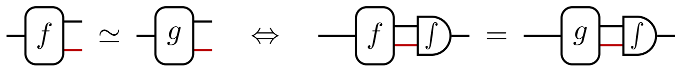
We now prove that the claimed properties indeed hold for our main example.

Proposition 2.7. *The integrator kernels $\int_Y : Y \otimes W \rightsquigarrow Y$ of Definition 2.4 form a supply of integrators for $W^r := (\overline{\mathbb{R}}_{\geq 0}, \cdot, 1)$ on* sKern.

Proof. The left-hand side of the monad unit law acts by $t(x) \mapsto w \cdot t(x) \mapsto 1 \cdot t(x) = t(x)$, which agrees with the identity. The monad action law follows since the left-hand side sends $t(x) \mapsto w_2 \cdot t(x) \mapsto w_1 \cdot w_2 \cdot t(x)$, which coincides with the action of the right-hand side. Turning to naturality, the left-hand side acts by $t(x) \mapsto w \cdot t(x) \mapsto \int_X w \cdot t(x)\, f$. The right maps $t(x) \mapsto \int_X t(x)\, f \mapsto w \cdot \int_X t(x)\, f$, which coincides by linearity of the integral. The compatibility with the monoidal product follows since both sides act by $t(x,y) \mapsto w_1 w_2 \cdot t(x,y)$. Finally, we note that compatibility with the monoidal unit is a special case of the algebra unit law. ■

2.1.4 Weight equivalence

Definition 2.8 (Weight equivalence)**.** Let W be a weights monoid in an SMC $\mathcal{C}$. We say that arrows $f, g \in \mathcal{C}_W(X, Y)$ are *weight equivalent* iff $\int_W f = \int_W g$. In this case, we write $f \simeq g$. In string diagrams, weight equivalence becomes



Remark 2.9. Weight equivalence is compatible with sequential and parallel composition since it is defined by the symmetric strict monoidal functor $\int_W$.

2.1.5 Computing with weighted kernels

We are now in the position to describe how weighted kernels can be used as a computational tool. Suppose we want to approximate an s-finite kernel $m\colon X \to Y$. Assume we can construct a weighted *probability* kernel $k\colon X \to Y \otimes W$ with $\int_W k = m$. The next section will show that this is always possible. Given any test function $t \in \mathfrak{M}^+(Y)$, we can write $m \fatsemi t = (\int_W k) \fatsemi t = k \fatsemi (\int_Y \fatsemi t)$. Hence, given independent samples $(y^{(i)}, w^{(i)}) \sim k(- \mid x)$, we can approximate $m \fatsemi t$ using the usual Monte Carlo estimator for $(\int_Y \fatsemi t)$:

$$(m \fatsemi t)(\bullet \mid x) \approx \frac{1}{N} \sum_i (\int_Y \fatsemi t)(y^{(i)}, w^{(i)}) = \frac{1}{N} \sum_i w^{(i)}\, t(y^{(i)}).$$

The approximation converges almost surely by the law of large numbers, showing that we can represent m using weighted samples from k.

In particular, if k is constructed by weighting a probability kernel $q\colon X \rightsquigarrow Y$ by a deterministic weight function $w\colon X \times Y \to W$, then the above recovers classic ***importance sampling***: We may draw samples from $k(- \mid x)$ by first sampling $y^{(i)} \sim q(- \mid x)$ and then computing the importance weight $w^{(i)} := w(x, y^{(i)})$. The resulting estimator for $m \fatsemi t$ thus becomes $\frac{1}{N} \sum_i w(x, y^{(i)})\, t(y^{(i)})$. More generally, the above method may be regarded as random-weight importance sampling.

Since $\int_W$ is a symmetric strict monoidal functor, this approach extends to diagrams of s-finite kernels. In particular, suppose $m\colon X \rightsquigarrow Y$ is given by a diagram of s-finite kernels $f_j\colon X_j \rightsquigarrow Y_j$ for which there are weighted probability kernels $k_j\colon X_j \rightsquigarrow Y_j \otimes W$ with $\int_W k_j = f_j$. By functoriality, composing the k_j according to the diagram gives a weighted probability kernel $k\colon X \rightsquigarrow Y \otimes W$ with $\int_W k = m$. We may thus again approximate $t \in \mathfrak{M}^+(Y)$ by the weighted average $\frac{1}{N} \sum_i w^{(i)}\, t(y^{(i)})$, where $(y^{(i)}, w^{(i)}) \sim k(- \mid x)$. Such samples may be drawn sequentially from the k_j by viewing them as a diagram in Stoch: We go through the diagram in topological order, sampling $(y_j, w_j) \sim k_j(- \mid x_j)$ in turn, treating the monoid multiplication as deterministic kernels. Hence, this method is a random-weight version of ***sequential importance sampling***.

2.1.6 Representing kernels by weighted probability kernels

In this section we show that any s-finite kernel can be lifted to a weighted probability kernel. Explicit lifts are easy to construct when the kernel in question has a density.

Example 2.10. Suppose $k\colon X \rightsquigarrow Y$ is given by a density $\ell_k\colon Y \times X \to \mathbb{R}_{\geq 0}$ with respect to a measure ν, that is $k(B \mid x) = \int_Y \ell_k(y \mid x)\,\nu(dy)$. Further suppose that $q\colon X \rightsquigarrow Y$ is a probability kernel with density ℓ_q with respect to ν with the property that $\ell_q = 0$ implies $\ell_k = 0$. Then we may weight q by $w := \ell_k/\ell_q$ to get a weighted probability kernel $\bar{k}\colon X \rightsquigarrow Y \otimes \overline{\mathbb{R}}_{\geq 0}$ defined on rectangles by $\bar{k}(A, B \mid x) := \int_A [\![w(y \mid x) \in B]\!]\, q(dy \mid x)$. On test functions, $\bar{k}$ acts by $t(y, v) \mapsto \int_Y t(y, w(y \mid x))\, q(dy \mid x)$. Note that this is a lift of k since $\int_W \bar{k}$ acts by

$$\begin{aligned} t(y) \mapsto w\, t(y) \mapsto & \int_Y w(y \mid x)\, t(y)\, q(dy \mid x) \\ &= \int_Y \frac{\ell_k(y \mid x)}{\ell_q(y \mid x)}\, t(y)\, \ell_q(y \mid x)\, \nu(dy) \\ &= \int_Y t(y)\, \ell_k(y \mid x)\, \nu(dy) = k \mathbin{⨾} t. \end{aligned}$$

This idea generalizes by using the Radon-Nikodym theorem for s-finite kernels (Vákár and Ong 2018).

Definition 2.11 (Absolute continuity for s-finite kernels)**.** Let m be a measure on a space X. We call $U \in \Sigma_X$ a

(a) *null set*, if $m(U) = 0$,
(b) 0-∞ *set*, if for all $V \in \Sigma_X$ with $V \subseteq U$, we have $m(V) \in \{0, \infty\}$.

Moreover, for another measure m' on X, we write

(a) $m' \ll m$, if every null set of m is a null set of m',
(b) $m' \overset{\infty}{\ll} m$, if $m' \ll m$ and every 0-∞ set of m is also a 0-∞ set of m'.

We use the same terminology for kernels $f\colon X \rightsquigarrow Y$, if the corresponding property holds for $f(- \mid x)$, for all $x \in X$.

Lemma 2.12. *For σ-finite measures, every 0-∞ set is a null set. Hence, $\ll$ and $\overset{\infty}{\ll}$ coincide.*

Proof. Let U be a 0-∞ set of a σ-finite measure m on X, and suppose $m(V) = \infty$ for a measurable $V \subseteq U$. Let $\{E_i\}_{i \in \mathbb{N}}$ be a measurable partition of X with $m(E_i) < \infty$. Then $\sum_i m(V \cap E_i) = m(V)$ implies $m(V \cap E_i) > 0$ for some i. But $V \cap E_i \subseteq U$ implies $m(V \cap E_i) = \infty$, since U is a 0-∞ set, which contradicts $m(V \cap E_i) \leq m(E_i) < \infty$. Therefore $m(V) = 0$. ∎

Theorem 2.13 (Radon-Nikodym, Vákár and Ong 2018, Thm. 10)**.** *Let $k, k'\colon X \rightsquigarrow Y$ be s-finite kernels with $k' \overset{\infty}{\ll} k$. Then there exists a function*

$$\frac{dk'}{dk}\colon Y \times X \to \overline{\mathbb{R}}_{\geq 0},$$

called the Radon-Nikodym derivative, *such that* $\frac{dk'}{dk}(- \mid x)$ *is measurable for all* $x \in X$, *and*

$$k'(A \mid x) = \int_A \frac{dk'}{dk}(y \mid x)\, k(dy \mid x).$$

When X *is countable and discrete, or* Y *is standard Borel,* $\frac{dk'}{dk}$ *can be made jointly measurable in* X *and* Y*. Finally, if* k' *is* σ*-finite, then* $\frac{dk'}{dk}$ *takes values in* $\mathbb{R}_{\geq 0}$.

Proposition 2.14. *Let* Y *be a standard Borel space. For every s-finite kernel* $k\colon X \rightsquigarrow Y$ *in* sKern *there exists a probability kernel* $\bar{k}\colon X \rightsquigarrow Y \otimes \overline{\mathbb{R}}_{\geq 0}$ *with* $\int_W \bar{k} = k$*. Moreover, when* k *is* σ*-finite, the weights can be chosen to lie in* $\mathbb{R}_{\geq 0}$.

Proof. Write $k = \sum_{i=1}^{\infty} f_i$ for finite kernels $f_i\colon X \rightsquigarrow Y$. Define a probability kernel

$$p(A \mid x) := \sum_{i=1}^{\infty} 2^{-i} \frac{f_i(A \mid x)}{f_i(Y \mid x)},$$

where we interpret $0/0$ as 1. Observe that $p(A \mid x) = 0$ implies $k(A \mid x) = 0$, and $p(A \mid x) = \infty$ implies $k(A \mid x) = \infty$. Hence, $k \overset{\infty}{\ll} p$ and Radon-Nikodym (Theorem 2.13) yields a measurable function $w\colon Y \otimes X \to \overline{\mathbb{R}}_{\geq 0}$ satisfying

$$k(A \mid x) = \int_A w(y \mid x)\, p(dy \mid x).$$

Using Fubini and integrating over delta measures,

$$\int_A w(y \mid x)\, p(dy \mid x) = \int_W w \int_A [\![w(y \mid x) \in dw]\!]\, p(dy \mid x),$$

showing $\int_W \bar{k} = k$ for the probability kernel $\bar{k}(A, B \mid x) := \int_A [\![w(y \mid x) \in B]\!]\, p(dy \mid x)$. When k is σ-finite, the Radon-Nikodym derivative w takes values in $\mathbb{R}_{\geq 0}$. ∎

2.2 Parallel weighted sampling

The sequential weighted sampling method described in Section 2.1.5 has the practical limitation that the weights tend exponentially towards zero as the number of sampling steps increases. Since low weight samples don't contribute towards estimating weighted averages, this means that most computation is wasted.

This problem can be addressed by ***resampling***: Instead of drawing samples on at a time, we maintain a population of weighted samples, called ***particles***. Periodically, we eliminate low-weight samples and duplicate high-weight samples in a consistent manner. The simplest way of doing so, called ***multinomial resampling***, independently draws new sample indices according to their weights. In the following, we extend the weighted sampling theory from above to arrays of particles.

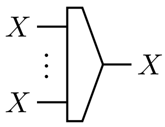
2.2.1 Mix

It is convenient to introduce the notation $x_{1:n} := (x_1, \ldots, x_n)$ for arrays.

Definition 2.15 (Mix)**.** For every space X and $n \in \mathbb{N}$, we define the *mixing kernel* $\mathsf{mix}\colon X^n \rightsquigarrow X$ by its action on $t \in \mathfrak{M}^+(X)$:

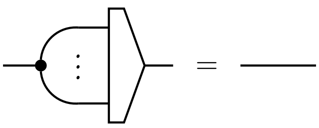
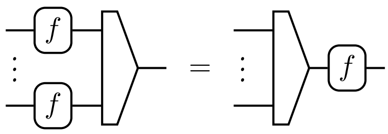
$$\mathsf{mix}(t(x) \mid x_{1:n}) := \frac{1}{n} \sum_{i=1}^{n} t(x_i).$$

We draw mixing as



Lemma 2.16. *Mixing satisfies the following properties:*

(a) *Copying then mixing is the identity.*
(b) *Mix is natural on probability kernels $f\colon X \rightsquigarrow Y$.*

Proof. For (a), the composite acts by $t(x) \mapsto \frac{1}{n}\sum_{i=1}^{n} t(x_i) \mapsto \frac{1}{n}\sum_{i=1}^{n} t(x) = t(x)$, which is the identity. For (b), the left-hand side acts by

$$t(y) \mapsto \bigotimes_{j=1}^{n} f\left(\frac{1}{n}\sum_{i=1}^{n} t(y_i) \mid x_i\right) = \frac{1}{n}\sum_{i=1}^{n}\bigotimes_{j=1}^{n} f(t(y_i) \mid x_i).$$

When f is a probability kernel,

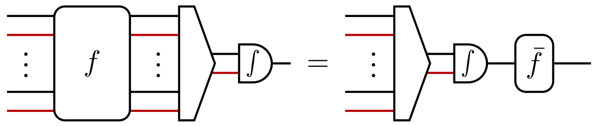
$$\bigotimes_{j=1}^{n} f(t(y_i) \mid x_i) = \int_{Y_1} \cdots \int_{Y_n} t(y_i)\, f(dy_1 \mid x_1) \cdots f(dy_n \mid x_n) = \int_{Y_i} t(y_i)\, f(dy_i \mid x_i).$$

Hence, the action reduces to $t(y) \mapsto \frac{1}{n}\sum_{i=1}^{n} f(t(y_i) \mid x_i)$, coinciding with right-hand side. ■

2.2.2 Category of weighted particles

Definition 2.17. A *particle transformer* $f\colon (X, m) \rightsquigarrow (Y, n)$ is a kernel of type $f\colon (X \otimes W)^m \rightsquigarrow (Y \otimes W)^n$. We think of f as accepting an array of m weighted samples and returning a random array of n weighted samples. Moreover, we use the convention that $(X, 0) = (X \otimes W)^0 = I$.

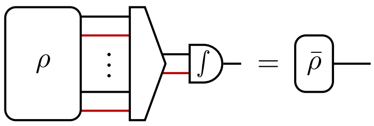
Definition 2.18 (Mixability)**.** For $m, n \geq 1$, a particle transformer $f\colon (X, m) \rightsquigarrow (Y, n)$ is called *mixable* if there exists a kernel $\bar{f}\colon X \rightsquigarrow Y$, called *the mean of* f, satisfying:

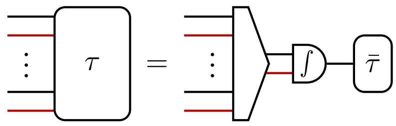


In terms of action on test functions $t \in \mathfrak{M}^+(Y)$, this is a kind of linearity condition.

$$\mathbb{E}_f\left(\frac{1}{n}\sum_{i=1}^{n} w_i\, t(y_i) \mid (x_j, v_j)_{j=1}^{m}\right) = \frac{1}{m}\sum_{j=1}^{m} v_j\, \mathbb{E}_{\bar{f}}(t(y) \mid x_j),$$

In case $n = 0$ or $m = 0$, we interpret taking the weighted average across no wires as id_I:





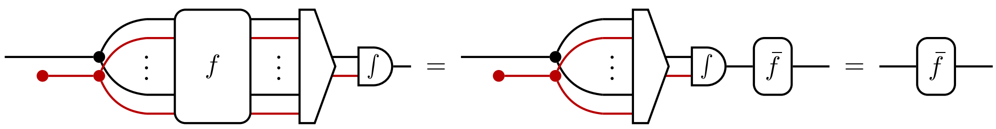
Hence, $\rho\colon I \rightsquigarrow (Y, n)$ is always mixable with $\mathbb{E}_{\bar{\rho}}(t(y)) := \mathbb{E}_\rho\left(\frac{1}{n}\sum_{i=1}^{n} w_i\, t(y_i)\right)$.

The equality $\mathbb{E}_{\bar{\rho}}(t(y)) = \mathbb{E}_\rho\left(\frac{1}{n}\sum_{i=1}^{n} w_i\, t(y_i)\right)$ tells us that if we compute the empirical weighted average $\frac{1}{n}\sum_{i=1}^{n} w_i\, t(y_i)$ over the output of ρ, the result will be the same as $\mathbb{E}_{\bar{\rho}}(t(y))$ in expectation. This result extends to all mixable particle transformers.

Lemma 2.19. *If $f\colon (X, m) \rightsquigarrow (Y, n)$ is mixable, then its mean is uniquely given by evaluating the average $\frac{1}{n}\sum_{i=1}^{n} t(y_i)$ at the input sequence $(x_j, v_j) := (x, 1)$, for $1 \leq j \leq m$:*

$$\mathbb{E}_{\bar{f}}(t(y) \mid x) = \mathbb{E}_f\left(\frac{1}{n}\sum_{i=1}^{n} w_i\, t(y_i) \mid (x, 1)_{j=1}^{m}\right).$$

Proof. Suppose $m, n \geq 1$. We pre-compose both sides of the defining property with $(\mathsf{id}_X \otimes e) \fatsemi \mathsf{cp}^m_{X \otimes W}$:



The final equality follows since $\mathsf{cp} \fatsemi \mathsf{mix} = \mathsf{id}$ (Lemma 2.16) and $(\mathsf{id}_X \otimes e) \fatsemi \int_X = \mathsf{id}_X$ (monad algebra unit law). Hence, the left-hand side provides a way to compute $\bar{f}$. On test functions it acts according to the claimed formula. The formula also holds for the cases $m = 0$ by disregarding the inputs, and for $n = 0$ by interpreting weighted averages over no indices as the identity. ■

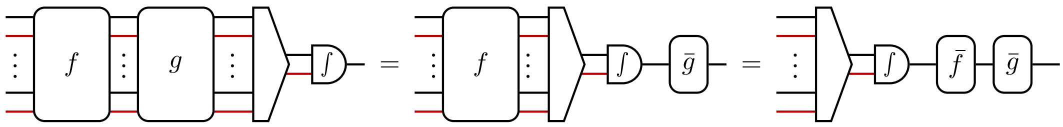
Particle transformers $f\colon (X, m) \rightsquigarrow (Y, n)$ and $g\colon (Y, n) \rightsquigarrow (Z, p)$ can be composed as kernels to obtain $(f \fatsemi g)\colon (X_m) \rightsquigarrow (Z, p)$.

Definition 2.20 (Category of particles)**.** We define the category of *weighted particles*, denoted Part, as follows:

(a) Its objects are pairs (X, n), where $X \in \mathsf{Ob}(\mathsf{sKern})$ and $n \in \mathbb{N}$.
(b) An arrow $f\colon (X, m) \rightsquigarrow (Y, n)$ is a mixable particle transformer.
(c) Composition and identities are inherited from sKern.

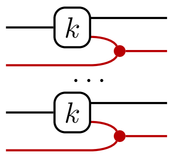
Proposition 2.21. *There is a functor $P\colon \mathsf{Part} \to \mathsf{sKern}$ that sends $f \mapsto \bar{f}$.*

Proof. It is immediate from the definition of mixability that $\overline{\mathsf{id}}_{(X\otimes W)^n} = \mathsf{id}_X$. Next, we check that $\overline{f \mathbin{⨾} g} = \bar{f} \mathbin{⨾} \bar{g}$. The case when $m, n \geq 1$ follows from the definition:



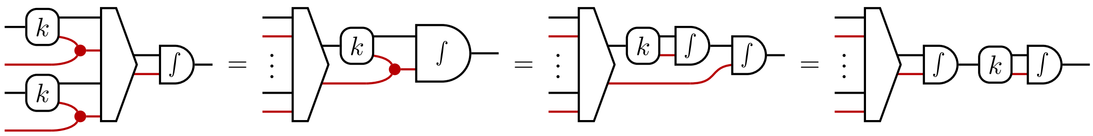
The argument continues to work using our conventions for $m = 0$ and $n = 0$. ∎

Proposition 2.22. *Given a weighted probability kernel $k\colon X \rightsquigarrow Y \otimes W$, we can lift it to a mixable arrow $k^n\colon (X, n) \rightsquigarrow (X, n)$ in* Part

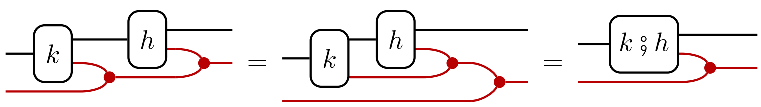


whose mean is $\int_W k$. This defines a functor $(-)^n\colon \mathsf{sKern}_W^{\mathsf{del}} \to \mathsf{Part}$.

Proof. We first argue that k^n is mixable with mean $\int_W k$ by the following calculation:



The first step uses the naturality of mix, the second the monad algebra action law, and the third the naturality of integrators. Next, we prove functoriality. The assignment preserves identities due to the monoid unit law. It preserves composition by monoid associativity and the definition of weighted kernel composition:



∎

Corollary 2.23 (Unbiasedness of particle estimates)**.** *The following diagram commutes:*

$$\begin{tikzcd}
\mathsf{sKern}_W^{\mathsf{del}} \arrow[rr, "(-)^n"] \arrow[dr, "\int_W"'] & & \mathsf{Part} \arrow[dl, "P"] \\
& \mathsf{sKern} &
\end{tikzcd}$$

This justifies particle algorithms in expectation: Suppose want to approximate an s-finite kernel $f\colon X \rightsquigarrow Y$. As discussed above, f lifts to a weighted probability kernel $k\colon X \rightsquigarrow Y \otimes \overline{\mathbb{R}}_{\geq 0}$ with $\int_W k = f$. By Corollary 2.23, we know $P(k^n) = \int_W k = f$. By Lemma 2.19, $\mathbb{E}_{P(k^n)}(t(y) \mid x) = \mathbb{E}_{k^n}(\frac{1}{n}\sum_i w_i\, t(y_i) \mid (x,1)_{j=1}^m)$. Hence, drawing a sample $(y_{1:n}, w_{1:n})$ and computing the weighted average $\frac{1}{n}\sum_i w_i\, t(y_i)$ coincides with $f(t(y))$ in expectation. Since P and $\int_W$ are functors, this transfers to any sequential decomposition of k.

2.2.3 Resampling

The following definition abstracts the key property which classic resampling methods have.

Definition 2.24. A mixable particle transformer $r\colon (X, m) \to (X, n)$ is called a *resampler* if its mean is the identity $\bar{r} = \mathsf{id}_X$.

Since $P(r) = \mathsf{id}_X$, the functoriality of P allows us to arbitrarily insert resampling steps at the level of particle transformers without affecting the mean.

Example 2.25. Multinomial resampling defines a particle transformer $r\colon (X, n) \to (X, n)$.

2.2.4 Single-run convergence

So far we have established that particle algorithms $\rho\colon I(X, n)$ are correct in expectation. In practice, however, we would like to run the algorithm only once with the assurance that $\hat{t}_n := \frac{1}{n}\sum_i w_i\, t(y_i)$ converges to $\mathbb{E}_\rho[t(y)]$ as $n \to \infty$. For this we must also control the variance of $\hat{t}_n$ and ensure that is decreases as the number of particles grows.

Recall from Theorem 1.3 that any kernel $f\colon X \to Y$ can be equivalently viewed as a monotone linear operator $\mathbb{E}_f\colon \mathfrak{M}^+(Y) \to \mathfrak{M}^+(X)$ satisfying monotone convergence. In particular, for any $t \in \mathfrak{M}^+(Y)$, we have $\mathbb{E}_f(t(y) \mid x) = \int_Y t(y)\, f(dy \mid x)$. Moreover, since kernel composition is defined by composing the kernel operators, we get the *law of iterated expectations* $\mathbb{E}_{f \fatsemi g} = \mathbb{E}_f \circ \mathbb{E}_g$ for free.

Definition 2.26. Let $f\colon X \rightsquigarrow Y$ be a kernel. The *conditional variance* of a test function $t \in \mathfrak{M}^+(Y)$ with respect to f is defined by

$$\mathrm{Var}_f\left[t(y) \mid x\right] := \mathbb{E}_f\left[t(y)^2 \mid x\right] - \mathbb{E}_f\left[t(y) \mid x\right]^2.$$

Remark 2.27. By linearity $\mathbb{E}_f$, one can show that $\mathrm{Var}_f\left[t(y) \mid x\right] = \mathbb{E}_f\left[(t(y) - m)^2 \mid x\right]$, where $m := \mathbb{E}_f\left[t(y) \mid x\right]$. In other words, the variance measures the mean squared error of $t(y)$ to $\mathbb{E}_f\left[t(y) \mid x\right]$.

Lemma 2.28. $\mathrm{Var}_f[\alpha\, t(y) \mid x] = \alpha^2\ \mathrm{Var}_f[t(y) \mid x]$.

Proposition 2.29 (Law of total variance)**.**

$$\mathrm{Var}_{f \fatsemi g}\left[t(z) \mid x\right] = \mathbb{E}_f\left[\mathrm{Var}_g\left[t(z) \mid y\right] \mid x\right] + \mathrm{Var}_f\left[\mathbb{E}_g\left[t(z) \mid y\right] \mid x\right].$$

Proof. We apply the definitions, the law of iterated expectations, and linearity to see that

$$\begin{aligned}\mathbb{E}_{f\,;\,g}[t(z)^2 \mid x] &= \mathbb{E}_f[\mathbb{E}_g[t(z)^2 \mid y] \mid x] \\ &= \mathbb{E}_f[\mathrm{Var}_g[t(z) \mid y] + \mathbb{E}_g[t(z) \mid y]^2 \mid x] \\ &= \mathbb{E}_f[\mathrm{Var}_g[t(z) \mid y] \mid x] + \mathbb{E}_f[\mathbb{E}_g[t(z) \mid y]^2 \mid x]\end{aligned}$$

Subtracting $\mathbb{E}_{f\,;\,g}[t(z) \mid x]^2 = \mathbb{E}_f[\mathbb{E}_g[t(z) \mid y] \mid x]^2$ yields the desired result. ■

Definition 2.30. Given a measurable space X, we denote the set of bounded measurable functions $s\colon X \to \mathbb{R}_{\geq 0}$ by $\mathfrak{M}_b^+(X)$.

In order to control the variance, we bound it by the supremum norm of the test function we are evaluating.

Definition 2.31 (Dispersion)**.** Let $\rho\colon I \to (X, n)$ be a particle transformer. We say that ρ has *dispersion* $d(\rho) \in \mathbb{R}_{\geq 0}$ if for any bounded measurable function $s \in \mathfrak{M}_b^+(X)$,

$$\mathrm{Var}_\rho\left[\frac{1}{n}\sum_k^n w_k\, s(x_k)\right] \leq \frac{d(\rho)}{n}\|s\|_\infty^2.$$

The concept of dispersion directly implies a ***weak law of large numbers*** via Chebychev's inequality. If μ is a measure on X and $\phi(x)$ a predicate, we abbreviate $\mu(\phi(x)) := \mu(\{x \in X : \phi(x)\})$. Then by definition $\mu(\phi(x)) = \mathbb{E}_\mu([\![\phi(x)]\!])$.

Proposition 2.32 (Markov / Chebychev inequalities)**.** *Let μ be a measure and $t \in \mathfrak{M}^+(X)$ measurable. For any $\epsilon > 0$,*

$$\mu(t(x) \geq \epsilon) \leq \frac{\mathbb{E}_\mu[t(x)]}{\epsilon}, \qquad \mu(|t(x) - \mathbb{E}_\mu[t(x)]| \geq \epsilon) \leq \frac{\mathrm{Var}_\mu[t(x)]}{\epsilon^2}$$

Proof. We note that since $t \geq 0$, we have $\epsilon\,[\![t(x) \leq \epsilon]\!] \leq t(x)$. Hence, by monotonicity and linearity of expectation

$$\epsilon\,\mu(t(x) \leq \epsilon) = \epsilon\,\mathbb{E}_\mu([\![t(x) \leq \epsilon]\!]) = \mathbb{E}_\mu(\epsilon\,[\![t(x) \leq \epsilon]\!]) \leq \mathbb{E}_\mu[t(x)].$$

Dividing by ϵ yields Markov's inequality. Chebychev's inequality follows by observing

$$\mu(|t(x) - \mathbb{E}_\mu[t(x)]| \geq \epsilon) = \mu((t(x) - \mathbb{E}_\mu[t(x)])^2 \geq \epsilon^2)$$

and applying Markov's inequality. ■

Definition 2.33 (Convergence in measure)**.** Let μ be a measure on X. We say that a sequence of measurable functions $t_n \in \mathfrak{M}^+(X)$ *converges in measure* to $t \in \mathfrak{M}^+(X)$ with respect to μ if for any $\epsilon > 0$,

$$\mu(|t_n(x) - t(x)| \geq \epsilon) \to 0$$

as $n \to \infty$. In this case we write $t_n(x) \xrightarrow{\mu} t(x)$.

Theorem 2.34 (Weak LLN)**.** *Let $\rho\colon I \to (X, n)$ be a particle transformer with dispersion d. Then for every bounded measurable function $s \in \mathfrak{M}_b^+(X)$ we have*

$$\frac{1}{n}\sum_{k}^{n} w_k\, s(x_k) \xrightarrow{\rho} \bar{\rho}(s(x)).$$

Proof. Fix $\epsilon > 0$ and recall from Lemma 2.19 that $\bar{\rho}(s(x)) = \mathbb{E}_\rho(\frac{1}{n}\sum_{k=1}^{n} w_k\, s(x_k))$. Hence, by Chebychev's inequality and the definition of dispersion

$$\rho\left(\left|\frac{1}{n}\sum_{k}^{n} w_k\, s(x_k) - \bar{\rho}(s(x))\right| \geq \epsilon\right) \leq \frac{1}{\epsilon^2}\operatorname{Var}_\rho\left[\frac{1}{n}\sum_{k}^{n} w_k\, s(x_k)\right] \leq \frac{d\,\|s\|_\infty^2}{n\epsilon^2}.$$

The latter converges to 0 as $n \to \infty$, proving convergence in measure. ∎

In order for composed particle transformers to be well behaved, we also need to control the second moment of the weights.

Definition 2.35 (Mean second moment)**.** Given a particle transformer $\rho\colon I \to (X, n)$, its *mean second moment* is $e(\rho) := \mathbb{E}_\rho\left[\frac{1}{n}\sum_{k=1}^{n} w_k^2\right]$.

Putting the above together yields a compositional stability criterion for particle transformers.

Definition 2.36 (Matrix stability)**.** A particle transformer $f\colon (X, n) \to (Y, m)$ is called *matrix-stable* if there exists a 2×2 $\mathbb{R}$-valued matrix M such that for every particle transformer $\rho\colon I \to (X, n)$ we have

$$\begin{bmatrix} d(\rho \mathbin{⨟} f) \\ e(\rho \mathbin{⨟} f) \end{bmatrix} \leq M \begin{bmatrix} d(\rho) \\ e(\rho) \end{bmatrix}.$$

Proposition 2.37. *Matrix-stable particle transformers are closed under composition.*

Proof. This follows immediately since

$$\begin{bmatrix} d(\rho \mathbin{⨟} f \mathbin{⨟} g) \\ e(\rho \mathbin{⨟} f \mathbin{⨟} g) \end{bmatrix} \leq M_g \begin{bmatrix} d(\rho \mathbin{⨟} f) \\ e(\rho \mathbin{⨟} f) \end{bmatrix} \leq M_f M_g \begin{bmatrix} d(\rho) \\ e(\rho) \end{bmatrix}.$$

∎

We conclude by showing that the typical particle transformers we use are matrix-stable.

Proposition 2.38. *Let $k\colon X \rightsquigarrow Y \otimes \overline{\mathbb{R}}_{\geq 0}$ be a weighted kernel satisfying*

(a) $A := \sup_{x \in X} \mathbb{E}_k[w \mid x] < \infty$,
(b) $B := \sup_{x \in X} \mathbb{E}_k[w^2 \mid x] < \infty$.

Then the particle transformer $k^n\colon (X, n) \to (Y, n)$ is matrix-stable with

$$M := \begin{bmatrix} A^2 & B \\ 0 & B \end{bmatrix}.$$

Proof. Let $\rho\colon I \to (X, n)$ be a particle transformer with dispersion $d(\rho)$. We first bound the dispersion $d(\rho \fatsemi k^n)$. By the law of total variance (Proposition 2.29), we can decompose the variance $\mathrm{Var}_{\rho \fatsemi k^n}\left[\frac{1}{n}\sum_i w_i\, s(y_i)\right] = V + E$, where

$$V := \mathrm{Var}_\rho\left[\mathbb{E}_{k^n}\left[\frac{1}{n}\sum_i w_i\, s(y_i) \mid (x_j, v_j)_{j=1}^n\right]\right], \quad E := \mathbb{E}_\rho\left[\mathrm{Var}_{k^n}\left[\frac{1}{n}\sum_i w_i\, s(y_i) \mid (x_j, v_j)_{j=1}^n\right]\right].$$

Since the k^n is mixable with mean $\int_W k$, and $\mathbb{E}_{\int_W k}[s(y) \mid x_j] = \mathbb{E}_k[v_j\, s(y_j) \mid x_j]$ we have

$$\mathbb{E}_{k^n}\left[\frac{1}{n}\sum_i w_i\, s(y_i) \mid (x_j, v_j)_{j=1}^n\right] = \frac{1}{n}\sum_j v_j\, \mathbb{E}_{\int_W k}[s(y_j) \mid x_j] = \frac{1}{n}\sum_j v_j\, \mathbb{E}_k[v_j\, s(y_j) \mid x_j].$$

By the definition of dispersion and using $\mathbb{E}_k[v_j\, s(y_j) \mid x_j] \le \|s\|_\infty \mathbb{E}_k[v_j \mid x_j] \le \|s\|_\infty A$,

$$V \le \frac{d(\rho)}{n}\|\mathbb{E}_k[v_j\, s(y_j) \mid x_j]\|_\infty^2 \le \frac{d(\rho)\, A^2\, \|s\|_\infty^2}{n}.$$

For the term E, we note that since k^n is an independent product over $(k \fatsemi (\mathsf{id}_Y \otimes m))$,

$$\mathrm{Var}_{k^n}\left[\frac{1}{n}\sum_i w_i\, s(y_i) \mid (x_j, v_j)_{j=1}^n\right] = \frac{1}{n^2}\sum_j \mathrm{Var}_{k \fatsemi (\mathsf{id}_Y \otimes m)}\left[w\, s(y) \mid x_j, v_j\right]$$

Since $(\mathsf{id}_Y \otimes m)$ is deterministic,

$$\mathrm{Var}_{k \fatsemi (\mathsf{id}_Y \otimes m)}\left[w\, s(y) \mid x_j, v_j\right] = \mathrm{Var}_k\left[v_j\, \tilde{w}\, s(y) \mid x_j\right] = v_j^2\ \mathrm{Var}_k\left[\tilde{w}\, s(y) \mid x_k\right]$$

Moreover, we can bound

$$\mathrm{Var}_k\left[\tilde{w}\, s(y) \mid x_j\right] \le \mathbb{E}_k[\tilde{w}^2\, s(y)^2 \mid x_j] \le \mathbb{E}_k[\tilde{w}^2\, s(y)^2 \mid x_j]\, \|s\|_\infty^2 \le B\, \|s\|_\infty^2.$$

Putting this together, we obtain

$$E \le \mathbb{E}_\rho\left[\frac{1}{n^2}\sum_j v_j^2\, B\, \|s\|_\infty^2\right] = \frac{B\, \|s\|_\infty^2}{n}\, \mathbb{E}_\rho\left[\frac{1}{n}\sum_j v_j^2\right] \le \frac{B\, \|s\|_\infty^2 e(\rho)}{n}.$$

Hence,

$$\mathrm{Var}_{\rho \fatsemi k^n}\left[\frac{1}{n}\sum_i w_i\, s(y_i)\right] \le \frac{A^2\, d(\rho) + B\, e(\rho)}{n}\, \|s\|_\infty^2$$

shows that $\rho \fatsemi k^n$ has dispersion bounded by $A^2\, d(\rho) + B\, e(\rho)$, as desired.

Next, we turn to the mean second moment. Using the definition, iterated expectations, and the structure of k^n, we compute

$$\begin{aligned} e(\rho \fatsemi k^n) &= \mathbb{E}_{\rho \fatsemi k^n}\left[\frac{1}{n}\sum_i w_i^2\right] = \mathbb{E}_\rho\left[\mathbb{E}_{k^n}\left[\frac{1}{n}\sum_i w_i^2 \mid (x_j, v_j)_{j=1}^n\right]\right] \\ &= \mathbb{E}_\rho\left[\frac{1}{n}\sum_j v_j^2\, \mathbb{E}_k\left[\tilde{w}^2 \mid x_j\right]\right] \le \mathbb{E}_\rho\left[\frac{1}{n}\sum_j v_j^2\, B\right] \\ &\le B\, \mathbb{E}_\rho\left[\frac{1}{n}\sum_j v_j^2\right] \le B\, e(\rho). \end{aligned}$$

This proves the claim about matrix stability. ∎

Proposition 2.39. *Multinomial resampling is* $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$*-stable.*

Proof. ∎

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